SUMMARY REPORT

SmartGridXAI: Explainable AI for Electricity Demand Prediction

1. INTRODUCTION

In today's current world of smart grid systems it is critical to accurately forecast the demand for electric energy so that energy can be managed effectively. Utilities companies rely heavily on being able to predict how much electric energy their customers will use, this will enable them to allocate appropriate resources for generating enough electric energy while minimising the waste, and ensure the electric grid continues functioning without any reliability issues.

Unfortunately, most of the conventional forecasting systems work as a black box and give the user a predicted value for electric energy demand without telling them why or how a particular electric energy demand has been predicted.

In this project we are proposing an explainable machine learning framework, called SmartGridXAI, for predicting electric energy demand based on data from smart meters and at the same time providing users with interpretable information on why energy demand happened. Our framework integrates traditional predictive modelling with explainable AI techniques in order to increase the understandability of our model for both energy analysts and decision makers.

2. DATASET

This project's dataset includes electric meter consumption records as well as the surrounding environment (temperature, humidity, wind speed) and the time of year (hour of the day and day of the week), all of which impact energy consumption.

Variables in the Dataset include:

- Temperature: It impacts how much energy is used for either heating or cooling.
- Humidity: It affects how much electricity is used.
- Wind Speed: It includes weather-related changes in energy use.
- Average Consumption: Electricity that has been consumed in the past.
- Hour: The time of day when electric energy is consumed.
- Day of Week: The pattern for consuming energy during the week.

Variable to Predict include:

- Electricity_Consumed: The amount of electricity that is consumed.

Each record in the dataset shows consumption of energy under different conditions and at different times.

3. METHODOLOGY

The project followed the standard data science process:

Preprocess Data → Feature Engineering → Train Model → Explain Model.

Data Preprocessing

The dataset had several preprocessing steps performed on it before using it for modelling:

- Converted timestamp to temporal features (Hour and Day of Week) for each record.
- Removed all irrelevant attributes from the dataset.
- Scaled features with the use of standardization.
- Split data into a training set and test set.

Machine Learning Model

A Random Forest Regressor was chosen for training and predicting electric consumption in this project because Random Forest:

- Effectively handles nonlinear relationships.
- Performs very well for tabular data.
- Provides information on variable importance.

Cross-validation was performed using RapidMiner on the dataset to validate that the model would perform well across all subsets of data being used to predict energy consumption.

4. MODEL EVALUATION

The determination of the performance of the model was made by evaluating its accuracy using several metrics, including:

- Mean Squared Error (MSE): The average of the squared differences between predicted values and actual values.
- Mean Absolute Error (MAE): The average of the absolute differences between predicted values and actual values.

- R^2 Score: A measure ranging from 0 to 1 that denotes the percentage of variation in the dependent variable that can be explained by independent variables.

The above-mentioned measures relate to how well the model can capture patterns in electricity usage and are helpful in determining the accuracy of prediction.

From the results produced by the above-mentioned measures, the model is capable of estimating electricity usage based on both environmental and temporal information.

5. EXPLAINABLE AI

The project employs two explainable AI techniques, including:

- SHAP (SHapley Additive Explanations) will provide a global view of how each variable contributes to the predicted value of the model. A SHAP summary plot provides an overall picture of which variables have the largest impact on electric usage.
- LIME (Local Interpretable Model-Agnostic Explanations) offers information to provide clarity about what features contributed to the prediction of electric usage for an individual case.

Both SHAP and LIME can be used as tools for stakeholders to understand how the model arrived at its conclusions and therefore provide a higher level of trust and transparency.

6. MAIN TAKEAWAYS

The output data from the modeling process gave the following important findings:

- The average past usage of electricity is the most significant predictive factor for the electrical load.
- Temperature/location conditions strongly impact the electrical load usage cycle and change the demand on the electrical grid.
- Time-based characteristics of electricity usage (like the time of day) are heavily influenced by human activity cycles throughout a 24-hour day; therefore the demand changes significantly between hour time periods.

7. RAPIDMINER INTEGRATION

RapidMiner was utilized to create a visual representation of a machine learning pipeline that also contained the following steps:

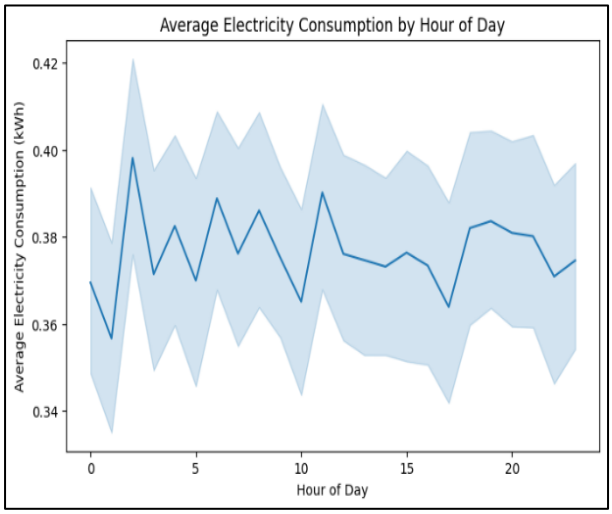
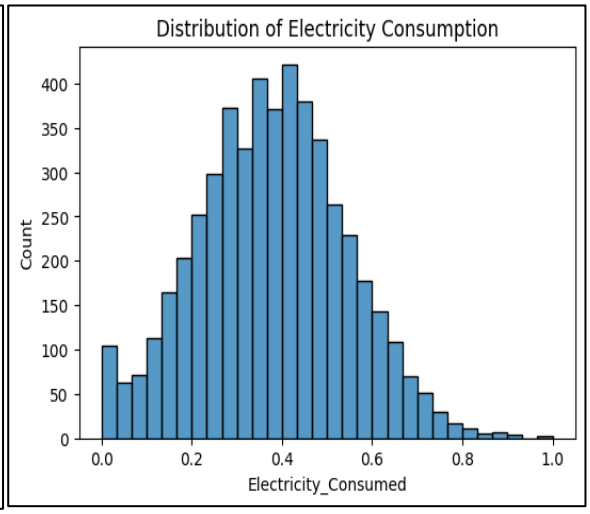
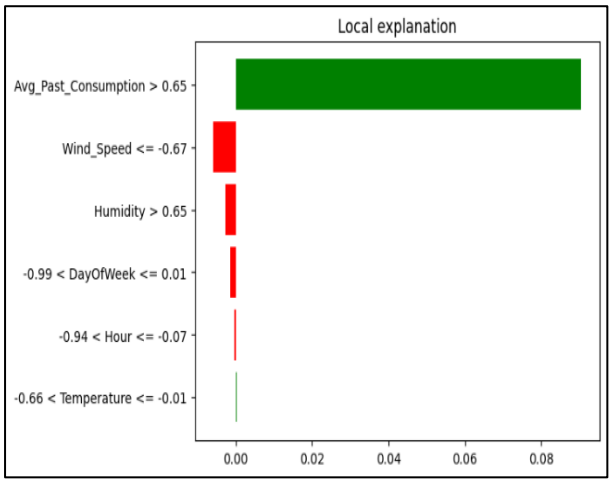
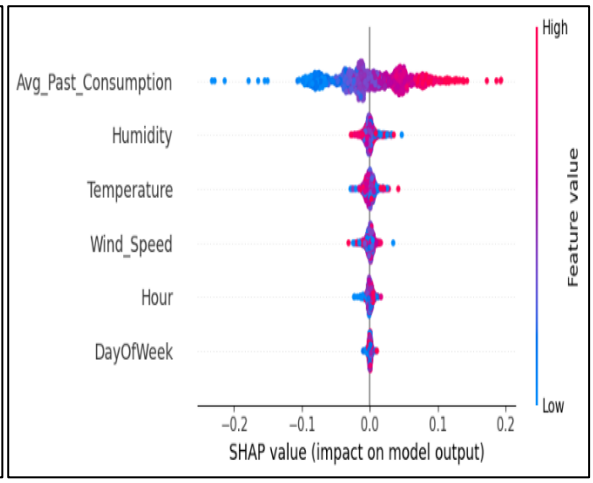
- Data Preprocessing and selection of features

- Normalization of features
- Model evaluation through cross-validation
- Random forest model training

The above described workflow is an example of the integration between data analytic tool sets into a predictive model pipeline.

8. CONCLUSION

The SmartGridXAI framework illustrates how machine learning with Explainable AI assists in forecasting electricity demand. By combining Random Forest modeling with SHAP & LIME explanations, the SmartGridXAI framework provides not just an estimate of electrical consumption but also an understanding of what drives electricity demand.

Such a methodology will aid in the management and operation of smart grids, planning of energy resources, and the data informed decisionmaking in today's energy systems.

9. VISUALIZATION SCREENSHOTS